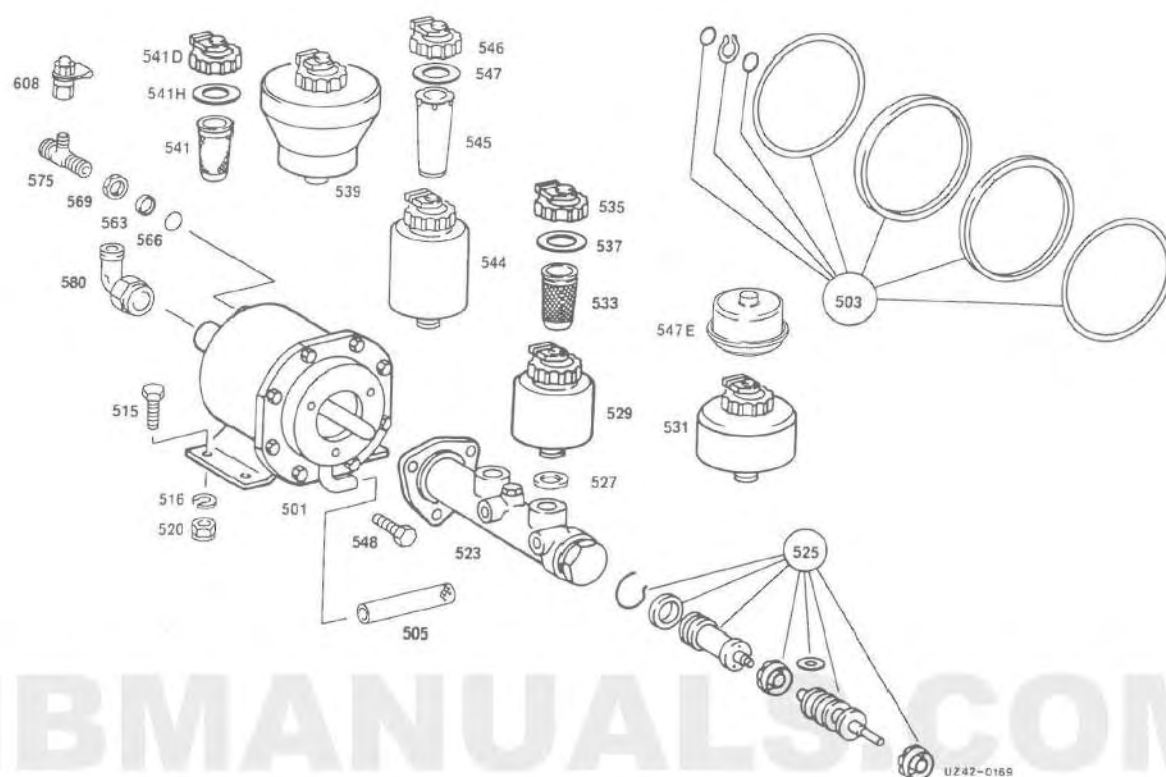


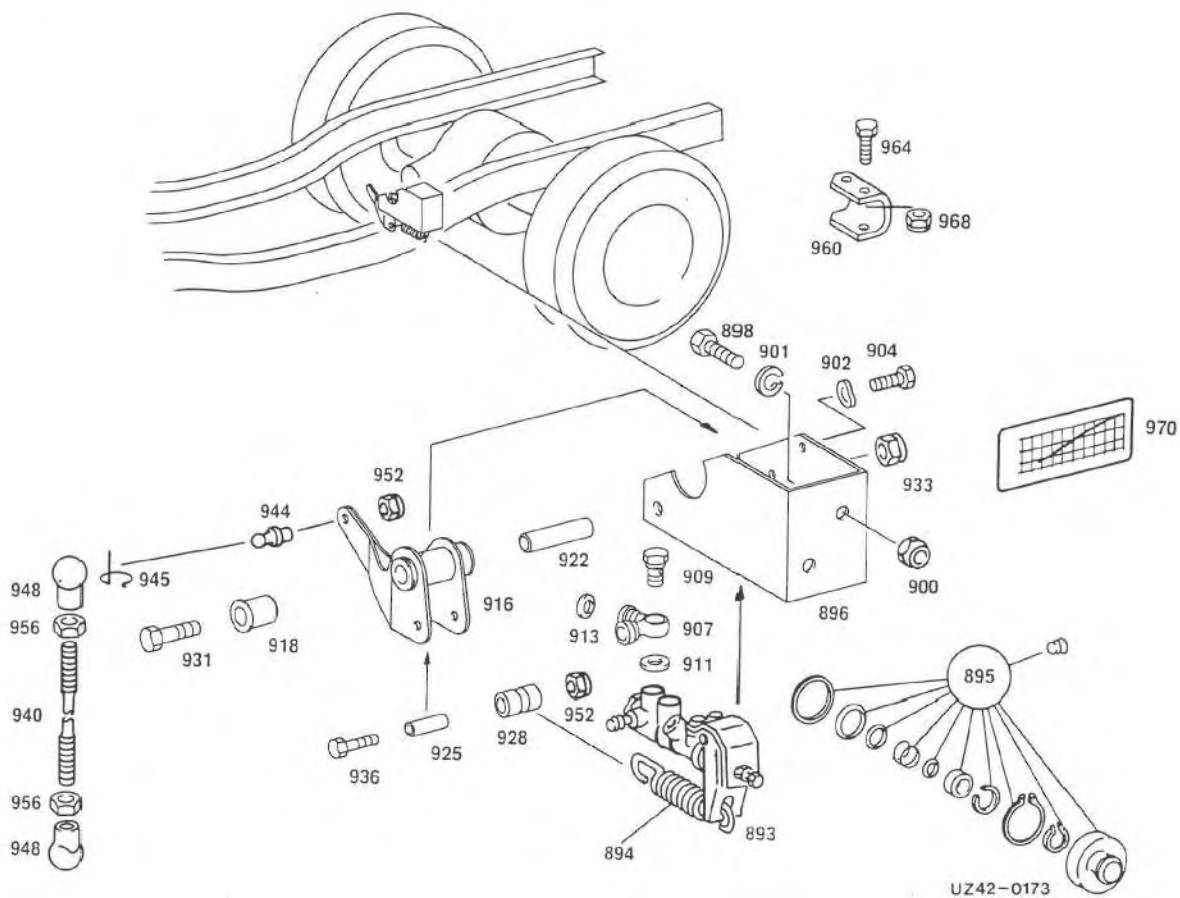
Exploded View



Preload Cylinder, Master Cylinder

501	Preload cylinder	541	Gauze filter
503	Repair kit	541D	Cap
505	Hose	541H	Sealing ring
515	Bolt	544	Reservoir 0,33 liter
516	Lock washer	545	Gauze filter
520	Nut	546	Cap
523	Tandem master cylinder	547	Sealing ring
525	Repair kit	547E	Protective cap
527	Sealing ring	548	Bolt
529	Reservoir 0,13 liter	563	Pressure spring
531	Reservoir 0,24 liter	566	Sealing ring
533	Gauze filter	569	Nut
535	Cap	575	Pipe socket
537	Sealing ring	580	Pipe socket
539	Reservoir 0,5 liter	608	Test connection

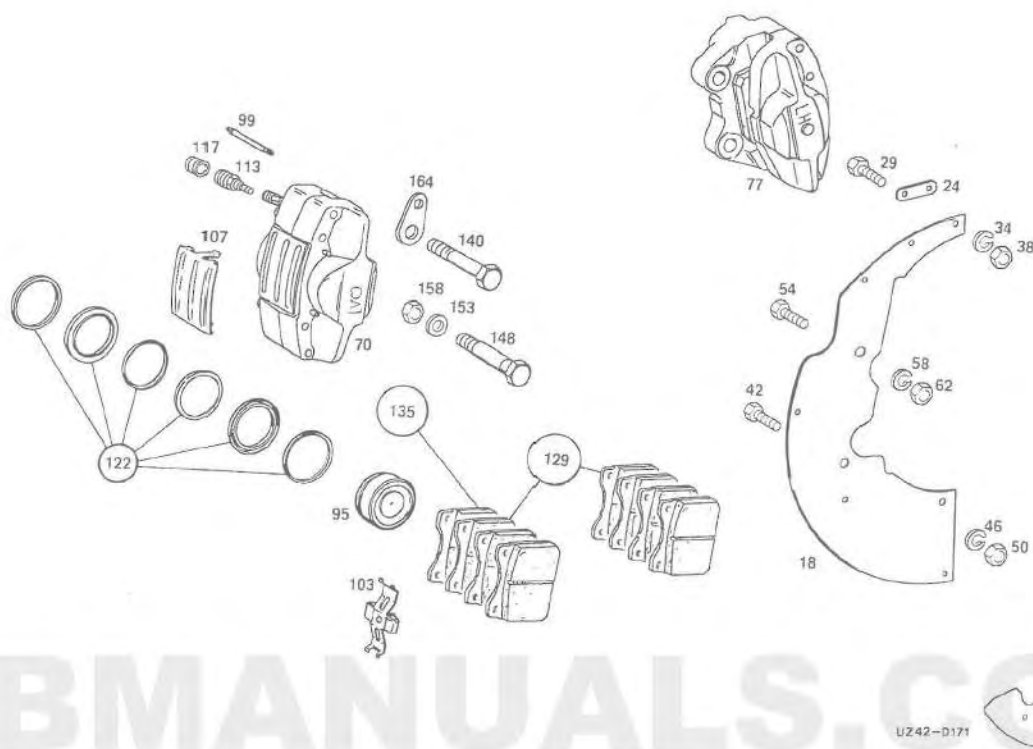
Exploded View



LB Modulator

- | | |
|-----------------------------|-----------------------|
| 93 Brake pressure regulator | 925 Bushing |
| 94 Spring | 928 Bearing |
| 95 Repair kit | 931 Bolt |
| 96 Bracket | 933 Nut |
| 98 Bolt | 936 Bolt |
| 10 Nut | 940 Pull rod |
| 11 Lock washer | 944 Ball pin |
| 12 Spring washer | 945 Clip |
| 14 Bolt | 948 Ball socket |
| 17 Distributor | 952 Nut |
| 19 Banjo bolt | 956 Nut |
| 11 Sealing ring | 960 Bracket |
| 13 Sealing ring | 964 Bolt |
| 16 Lever | 968 Nut |
| 18 Spacer tube | 970 Information plate |
| 22 Bushing | (ALB adjustment) |

Exploded View



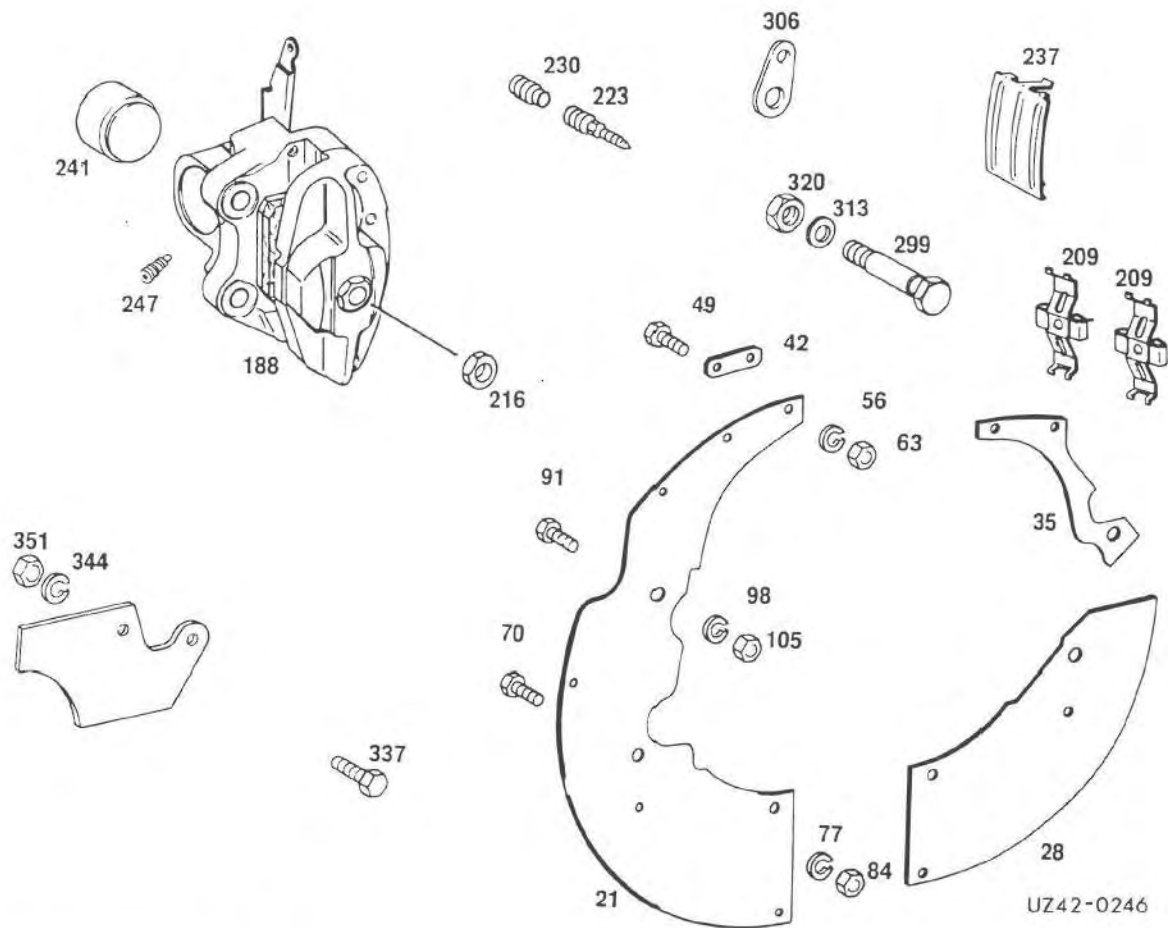
Brake Components, Front Axle

- | | |
|------------------------------|-----------------------------|
| 6 Backplate | 77 Fixed caliper, rear left |
| 12 Bolt | 95 Piston |
| 18 Backplate | 99 Pin |
| 24 Bracket | 103 Spring |
| 29 Bolt | 107 Guard plate |
| 34 Lock washer | 113 Bleeder valve |
| 38 Nut | 117 Dust cap |
| 42 Bolt | 122 Repair kit |
| 46 Lock washer | 129 Repair kit 8 pads |
| 50 Nut | 135 Repair kit 4 pads |
| 54 Bolt | 140 Bolt |
| 58 Lock washer | 148 Bolt |
| 62 Nut | 153 Lock washer |
| 70 Fixed caliper, front left | 158 Nut |
| | 164 Retainer |

42.11 General

747.111/113

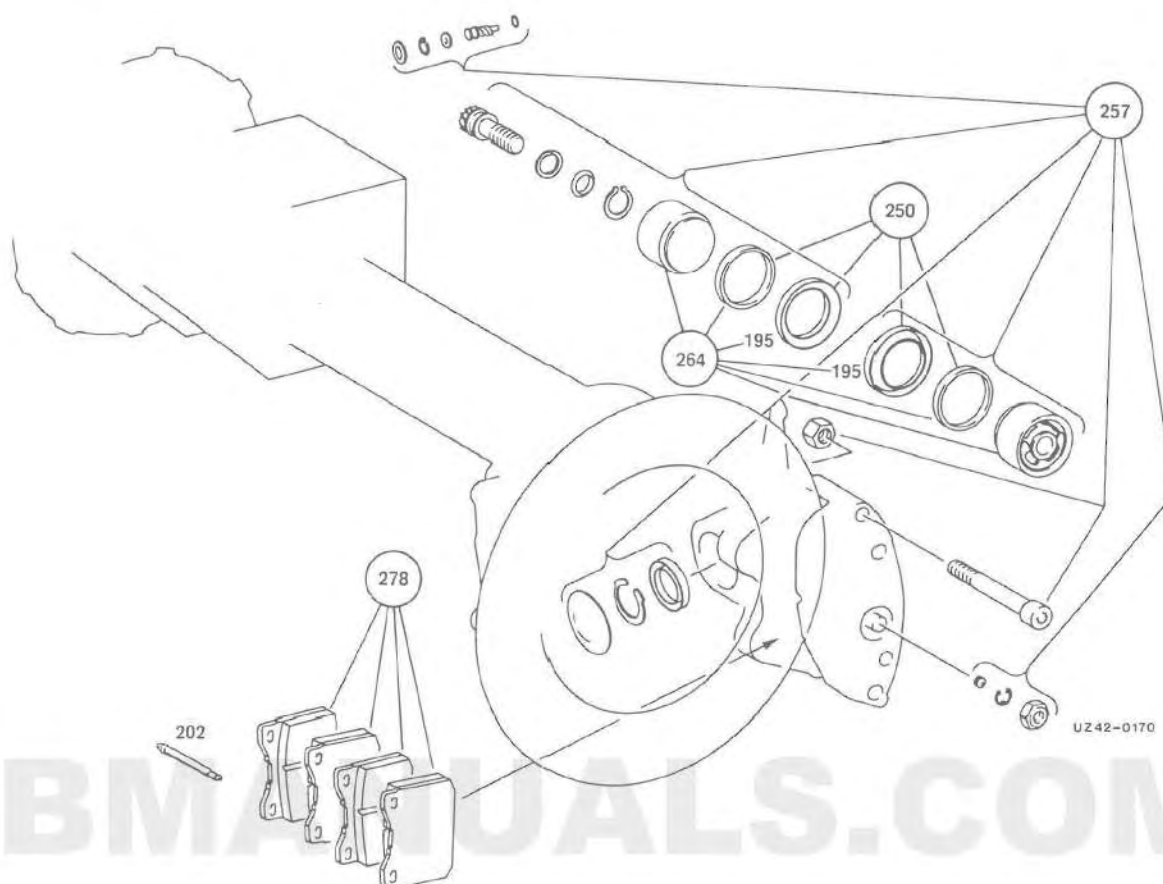
Exploded View



Brake Components, Rear Axle

21 Backplate	200 Spring
22 Backplate	216 Nut
28 Backplate, rear	223 Bleeder valve
35 Backplate, top	230 Dust cap
42 Bracket	237 Guard plate
49 Bolts	241 Piston
56 Lock washer	247 Pin
63 Nut	299 Bolts
70 Bolts	306 Retainer
77 Lock washer	313 Lock washer
84 Nut	320 Nuts
91 Bolt	337 Bolts
98 Lock washer	344 Lock washer
105 Nut	351 Nut
188 Brake caliper	

Exploded View

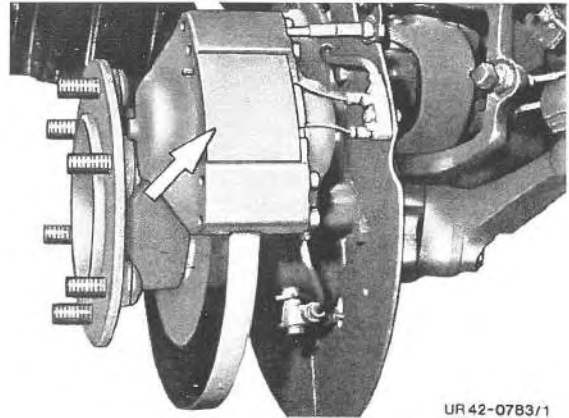


Brake Components, Rear Axle

- 195 Dust cap
- 202 Pin
- 250 Repair kit, sealing ring
- 257 Repair kit, readjustment
- 264 Repair kit, piston
- 278 Brake pad repair kit

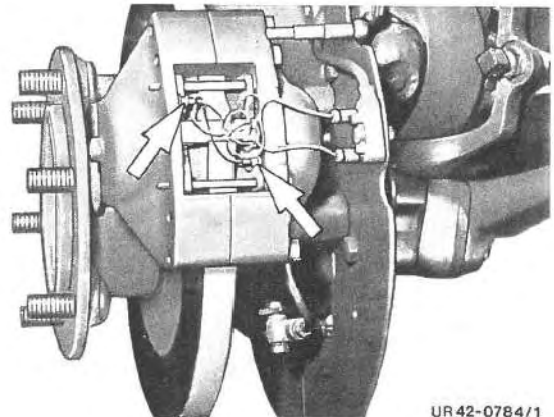
Removal

- 1 Remove wheel.
- 2 Remove cover plate, valid for brake pad wear indicator.



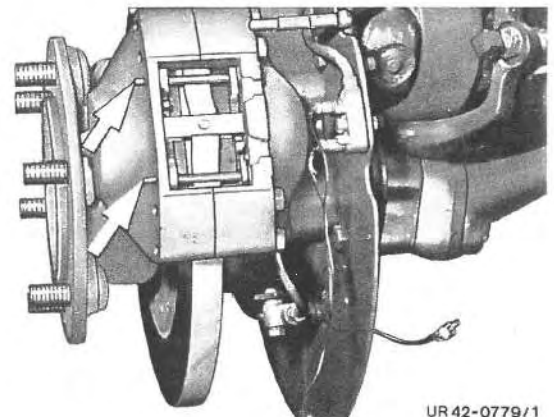
UR 42-0783/1

- 3 Detach cable, remove brake pads
Valid for brake lining wear indicator.



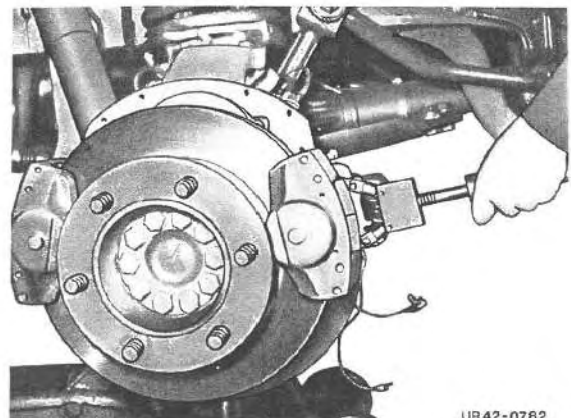
UR 42-0784/1

- 4 Drive out retaining pins and remove spring clip.



UR 42-0779/1

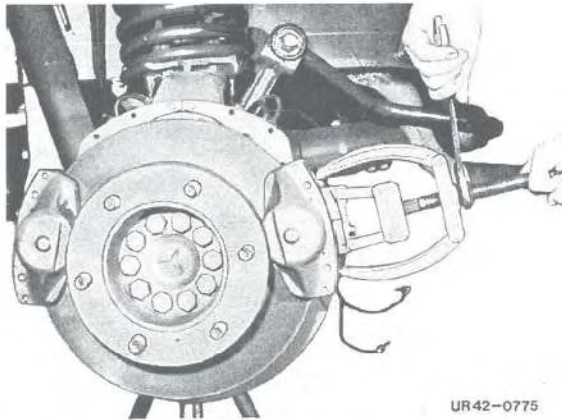
- 5 Remove brake pads using special tool No. 7.



UR 42-0762

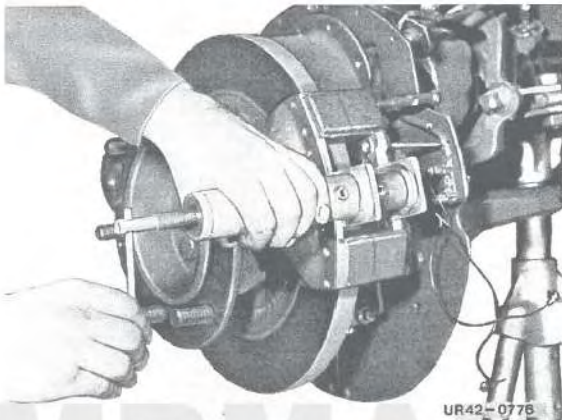
42.11 Removal and Installation of Brake Pads in Front Axle Fixed Caliper

737.111



6 If brake pads are seized, use special tool No. 6.

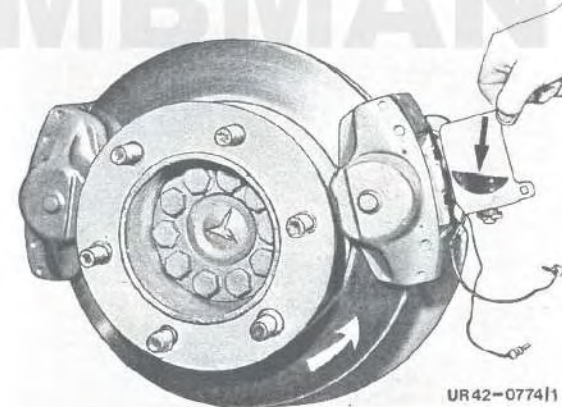
7 Remove backplate from shaft.



8 Using special tool No. 12, move piston in initial position.

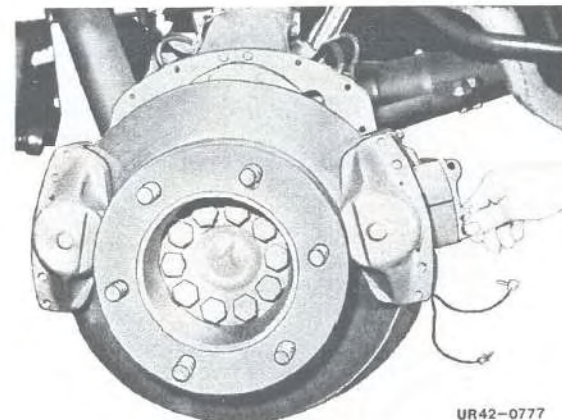
9 Clean fixed caliper shaft, brake disk and all removed parts, check and renew as required.

10 Clean oily brake disk with special cleaner No. 4, establish cause and correct.



Installation

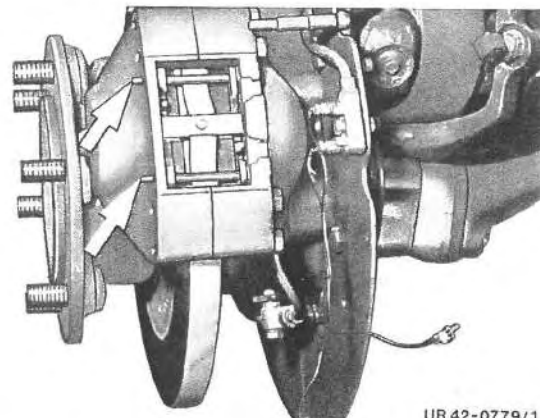
1 Fit backplate with opening facing disk run in.



2 Install brake pads.

Note: Only exchange brake pads in complete sets on one axle.

- 3 Fit spring clip and drive in retaining pins.



UR 42-0779/1

- 4 Install cable and cover plate. Valid for brake pad wear indicator.

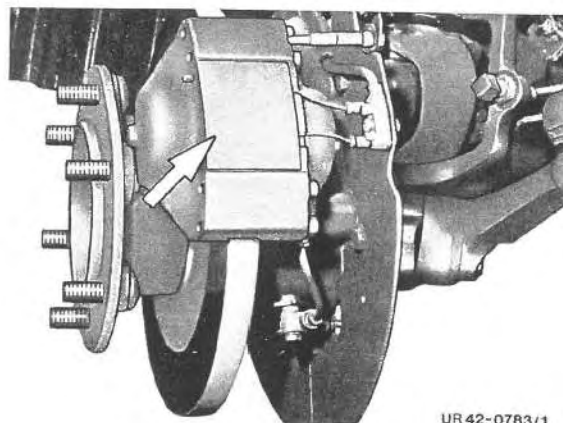
- 5 Fit wheel.

Tightening torque, see 1.3/1.

Note: Retighten wheel nuts after 50 km.

- 6 Actuate service brake several times and check that the wheel moves freely.

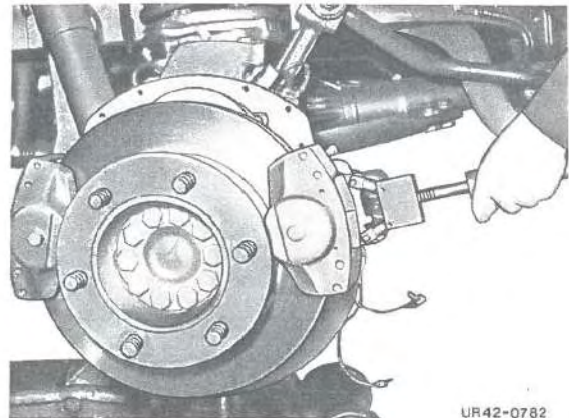
- 7 Check brake fluid level and top up if necessary.



UR 42-0783/1

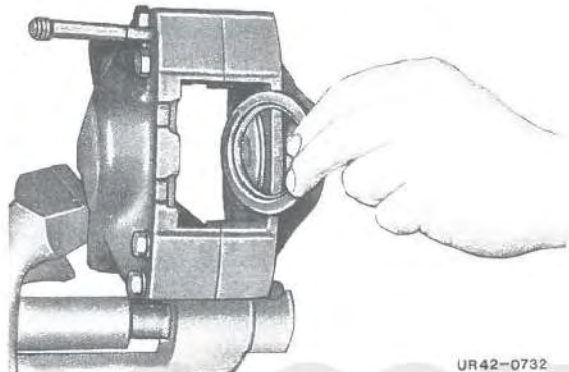
Removal

- 1 Remove brake pads, see 2.1/1.
- 2 Remove fixed caliper, see 33.6-6.1/1.



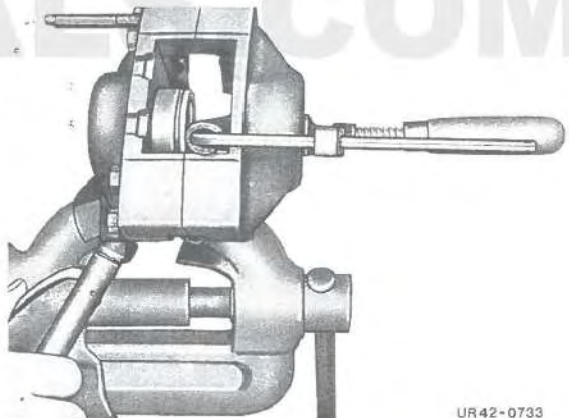
UR42-0782

- 3 Remove dust caps.



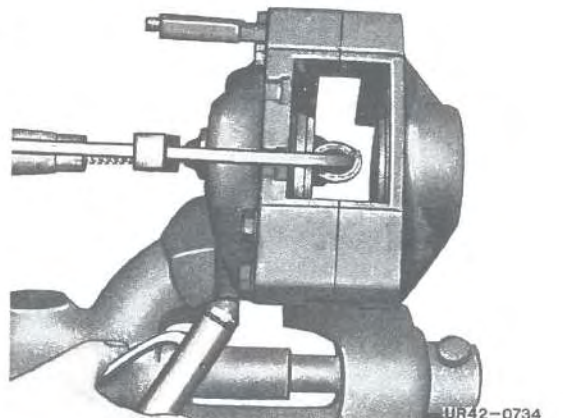
UR42-0732

- 4 Hold one piston and press out the free piston using compressed air.



UR42-0733

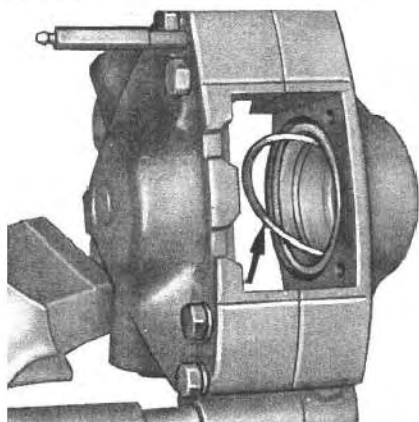
- 5 Close off cylinder bore using special tool No. 3 and press out the second piston.



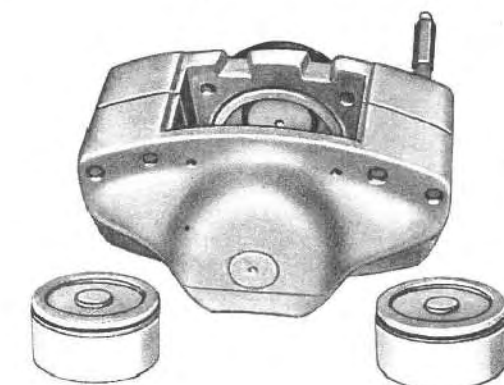
UR42-0734

2.11 Removal and Installation of Sealing Rings in Front Axle Fixed Caliper

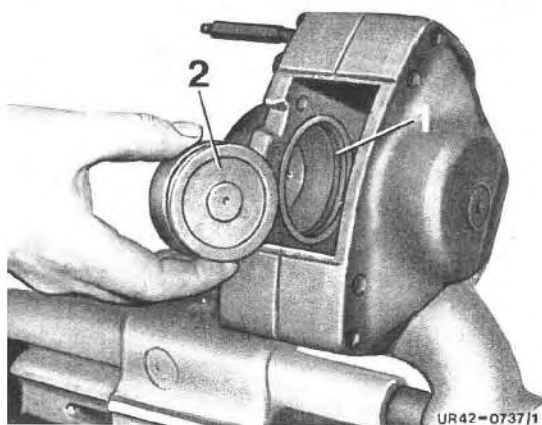
37.111



6 Remove sealing rings.

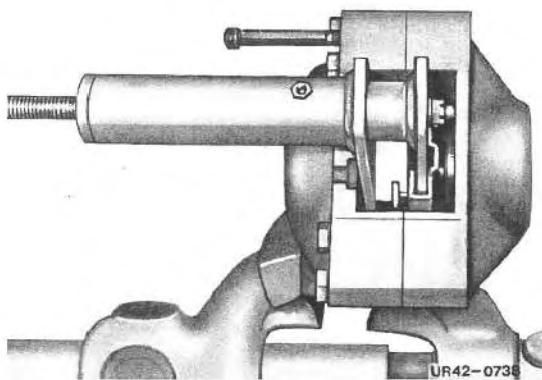


7 Clean fixed caliper and all removed parts, check and renew if necessary.



Installation

1 Coat sealing rings (1) and piston (2) with paste No. 2 and install.



2 Half fill the dust caps with paste No. 2 and press on to the shoulder in the cylinder bore using special tools No. 8, No. 11 and No. 12.

3 Attach fixed caliper, see 33.6-6.1/10.

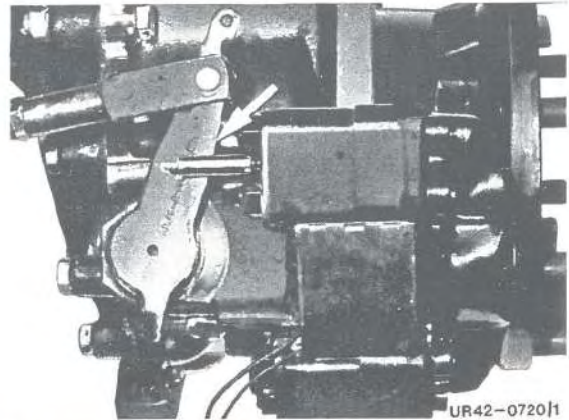
4 Install brake pads, see 2.1/2.

5 Bleed brake system, see 5.1/1.

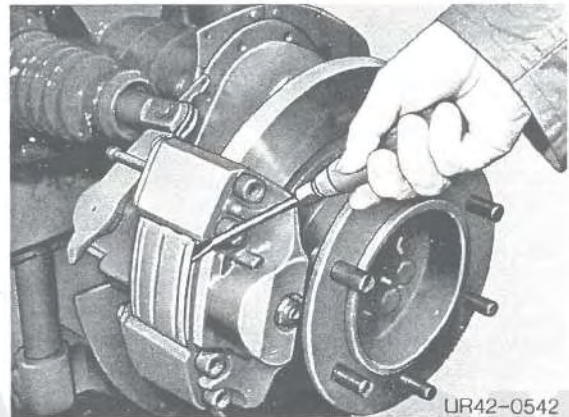
Removal

- 1 Detach wheel.
- 2 Release parking brake.

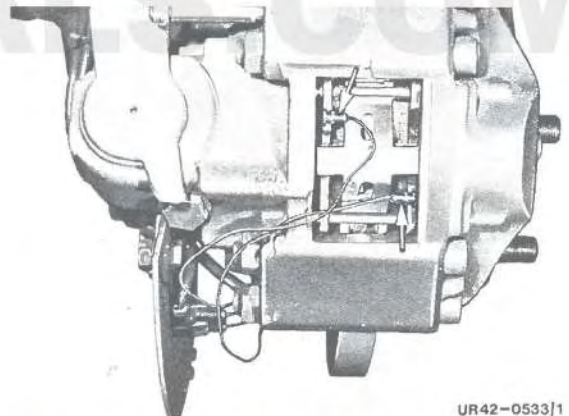
Note: Actuating lever must rest on stop.



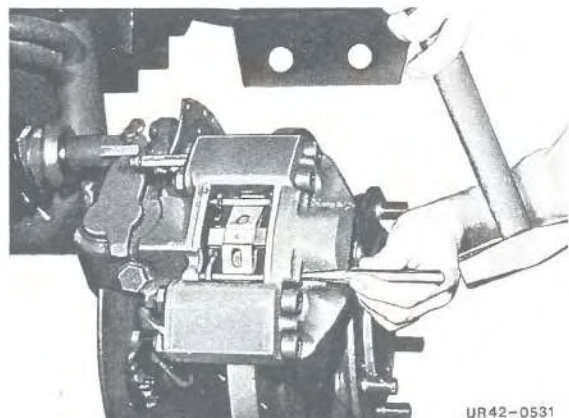
- 3 Remove guard plate.
Valid for brake pad wear indicator.



- 4 Detach cable from brake pads.
Valid for brake lining wear indicator.

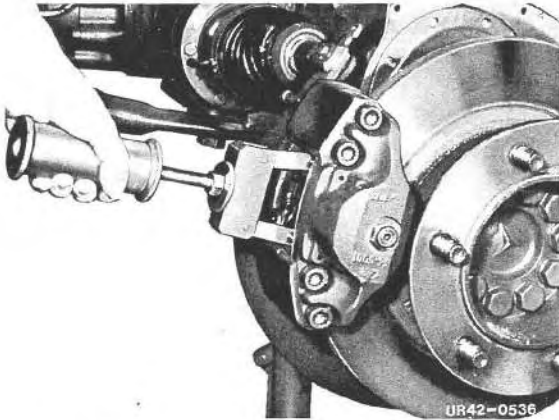


- 5 Drive out retaining pins and remove spring clip.

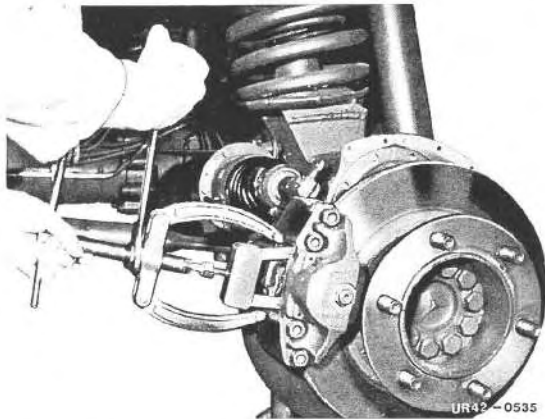


42.11 Removal and Installation of Brake Pads in Rear Axle Brake Caliper

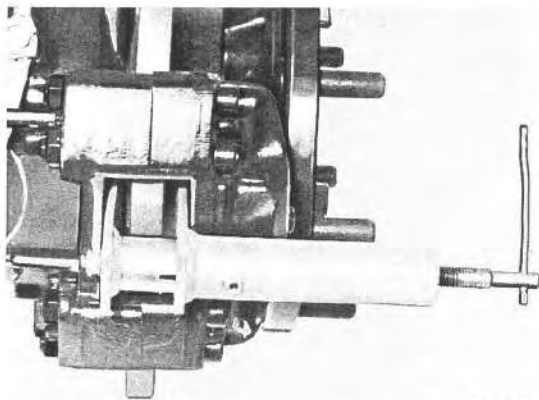
747.111/113



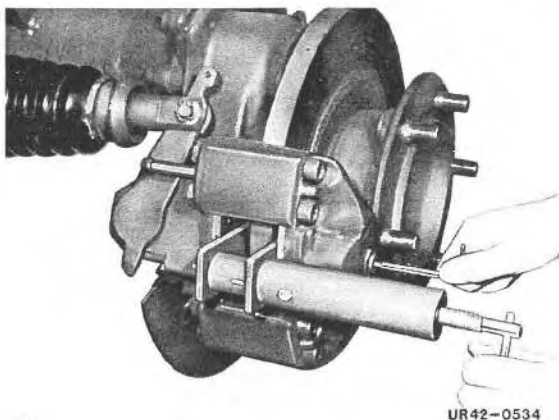
6 Withdraw brake pads using special tool No. 7.



7 If brake pads are stiff, use special tool No. 6.



8 Slightly tension piston using special tool No. 12.

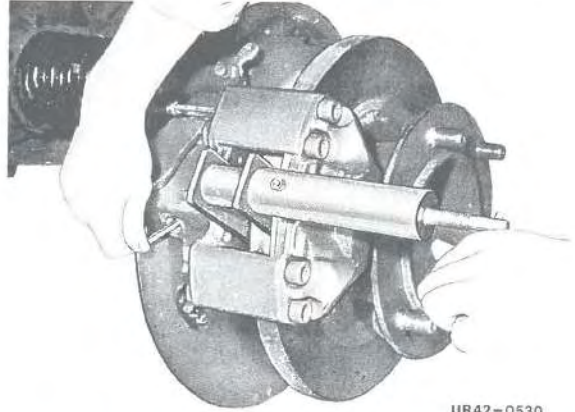


9 Loosen counternut and move piston on cover side to initial position by turning the adjusting screw. Direction of rotation, see 1.5/1.

Note: Maintain pressure on piston.

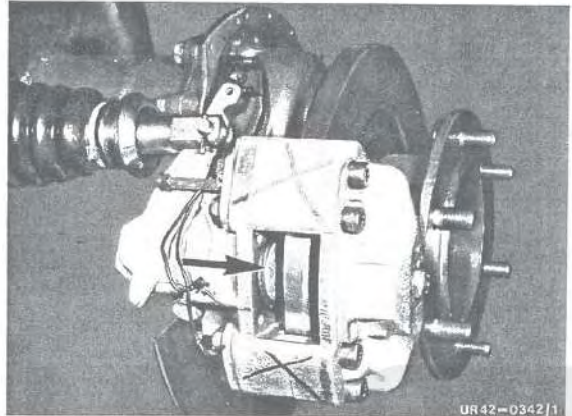
10 Unscrew screw plug and move piston on flange side to its initial position by turning the adjusting screw. Direction of rotation see 1.5/1.

Note: Maintain pressure applied to piston.



11 Clean shaft of fixed caliper, brake disk and all removed parts, check and renew if necessary.

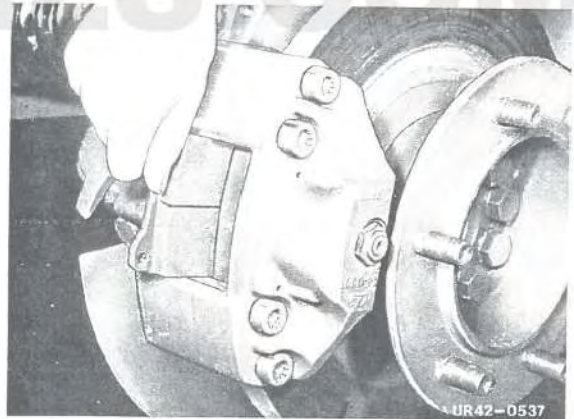
12 Clean oily brake disks with special cleaner No. 4, establish cause and correct.



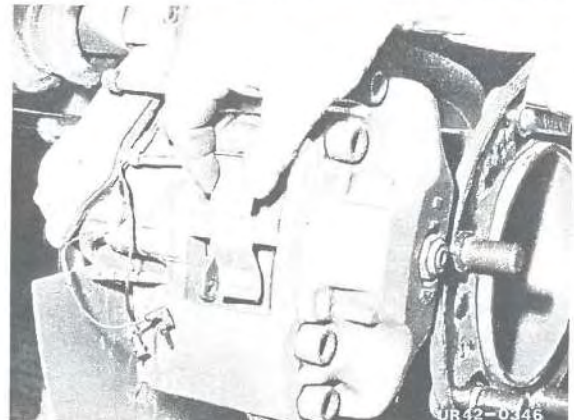
Installation

1 Install brake pads.

Note: Fit brake pads only in sets on one axle.

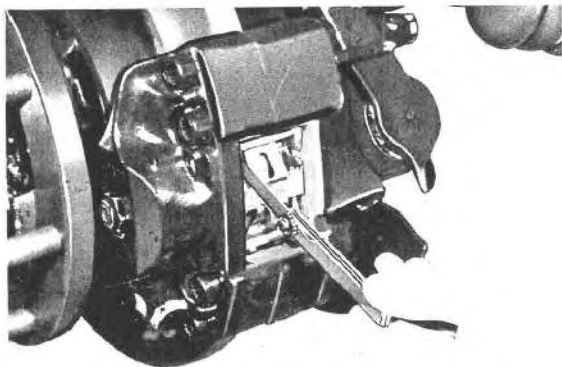


2 Fit spring clip and drive in retaining pins.



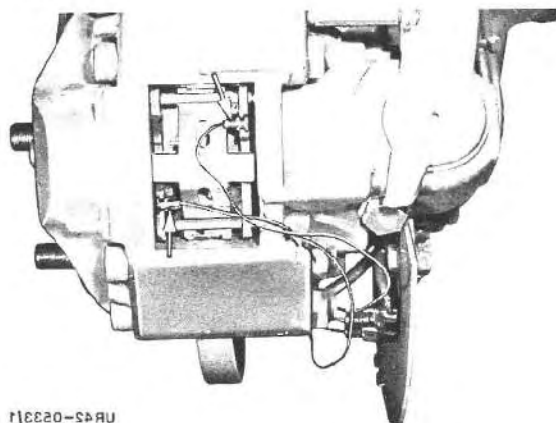
42.11 Removal and Installation of Brake Pads in Rear Axle Fixed Caliper

747.111/113



3. Adjust clearance, see 3.3/1.

NR45-0331



NR45-02331

4. Attach cable and guard plate.
Valid for brake pad wear indicator.

5. Attach wheel.
Tightening torque, see 1.3/1.

Note: Retighten wheel nuts after 50 km.

6. Actuate service brake several times and check that wheel moves freely.

7. Check brake fluid level and top up if necessary.

Disassembly

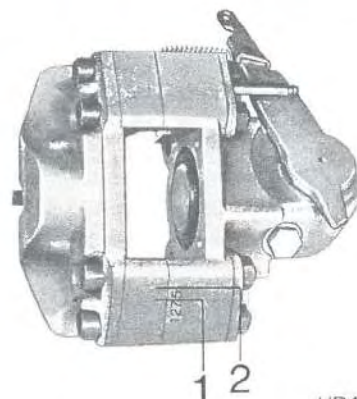
Note: Fixed calipers should be disassembled only by technicians in authorized workshops.

These repair instructions apply solely to fixed calipers dated by the manufacturer as of 3.11.75. The manufacturer's date is stamped in the upper side of the caliper flange indicating the consecutive day of the year (1) and the end of year digits (2).

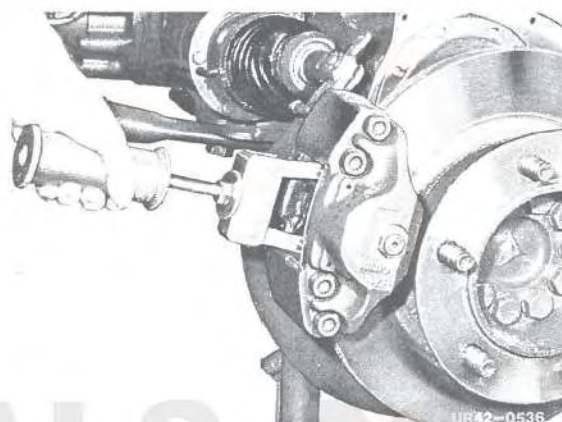
Fixed combination calipers manufactured prior to this date must be replaced by new calipers.

The item numbers () refer to page 1.4/1.

- 1 Remove brake pads, see 3.1/1.
- 2 Detach fixed caliper, see 35.6-6.1/2.

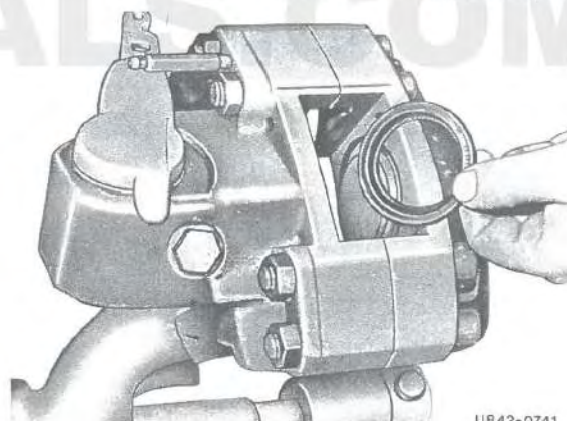


UR42-0437/1



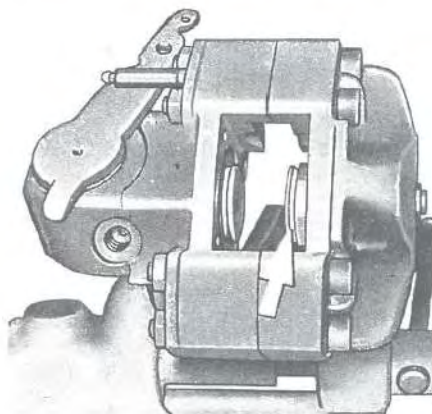
UR42-0536

- 3 Remove dust caps (10).



UR42-0741

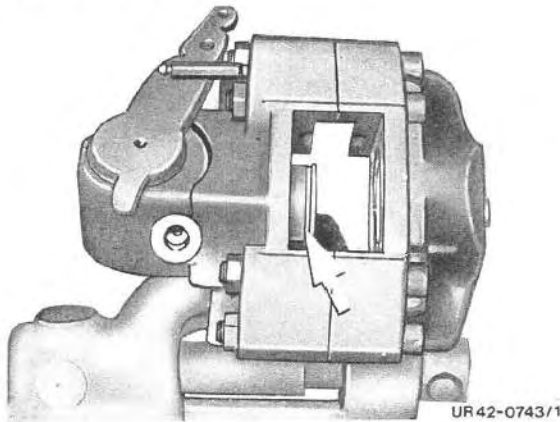
- 4 Remove piston (6) on the cover side.



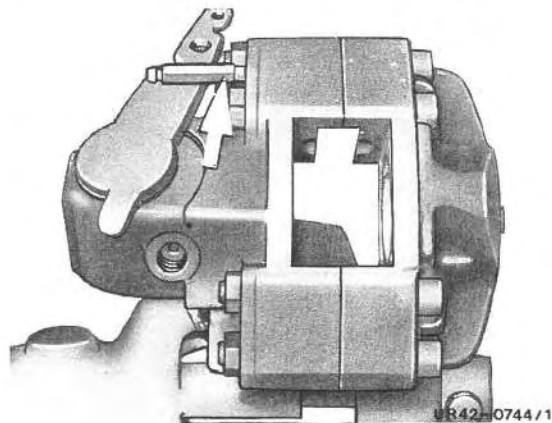
UR42-0742/1

42.11 Disassembly and Assembly of Rear Axle Fixed Caliper

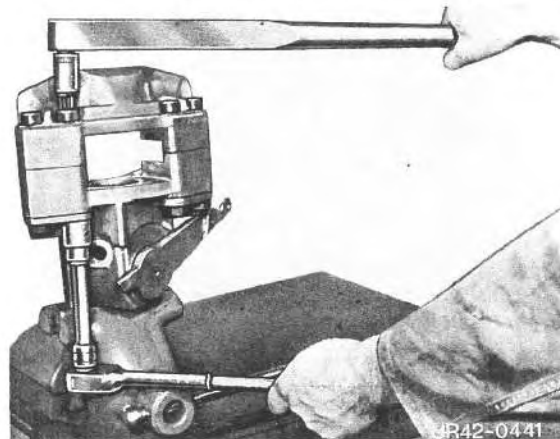
747.111/113



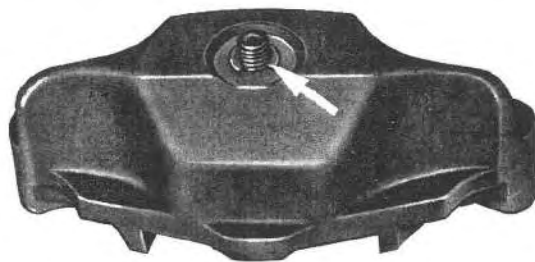
- 5 Remove piston (6) on flange side.



- 6 Unscrew bleeder valve.

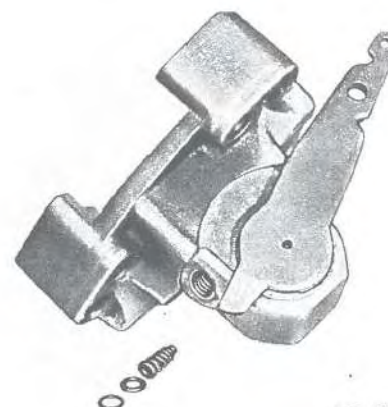


- 7 Attach fixed caliper to special tool No. 15, unscrew securing bolts with special tool No. 13 and separate fixed calipers.



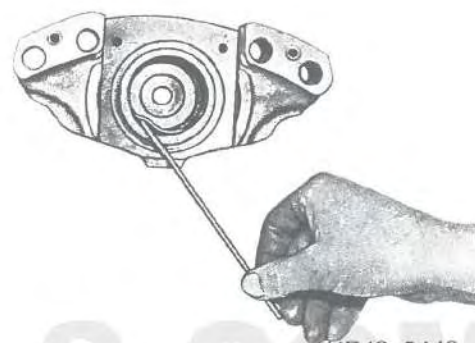
- 8 Release adjustment spindle (5) and remove.

9 Release restoring shaft (23) and remove.



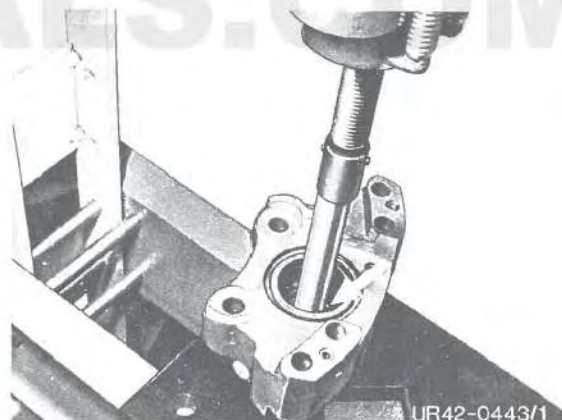
UR42-0449

10 Remove sealing rings (7).



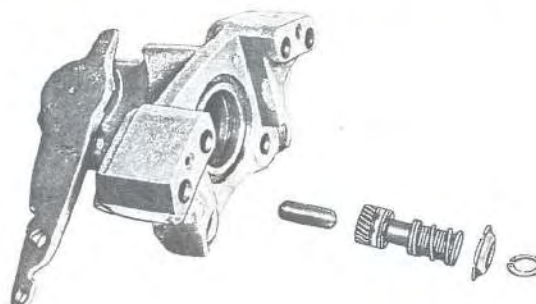
UR42-0442

11 Press special tool No. 14 on to spring plate (11) and remove retaining ring (12).



UR42-0443/1

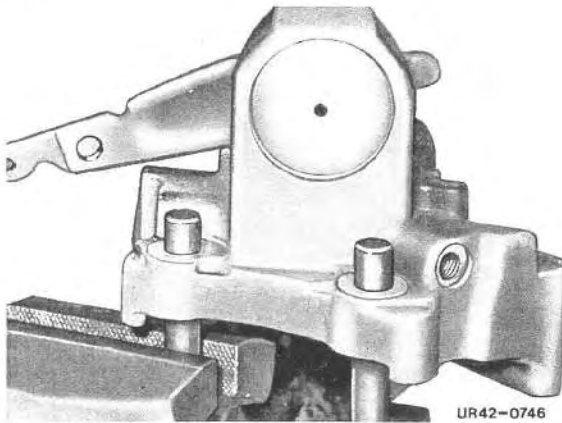
12 Remove spring plate (11), pressure spring (14), drive spindle (17) and plunger (24).



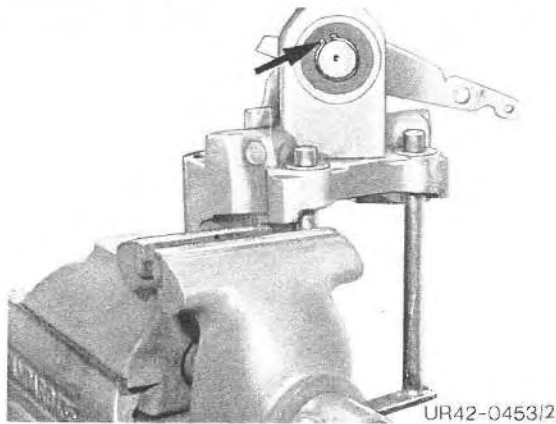
UR42-0450

42.11 Disassembly and Assembly of Rear Axle Fixed Caliper

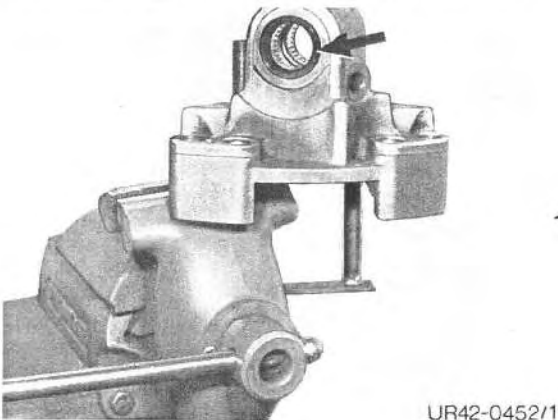
747.111/113



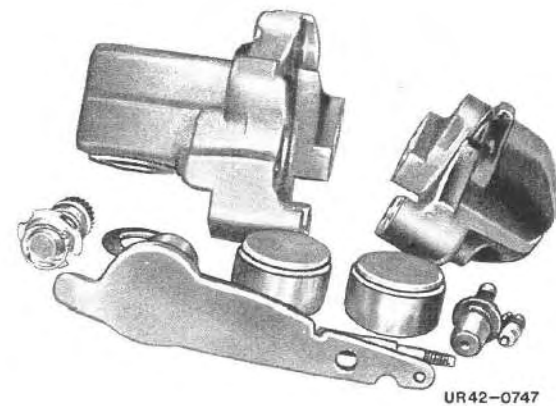
13 Drill open end plate (26) and remove.



14 Detach actuating lever (25) and support ring (28).



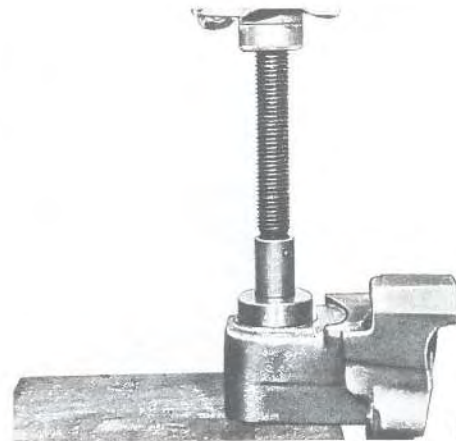
15 Remove sealing ring (29).



16 Clean all parts with cleaning agent No. 1, check and renew if necessary.

Assembly

1 Coat sealing ring (29) with paste No. 2 and press in using special tool No. 17.



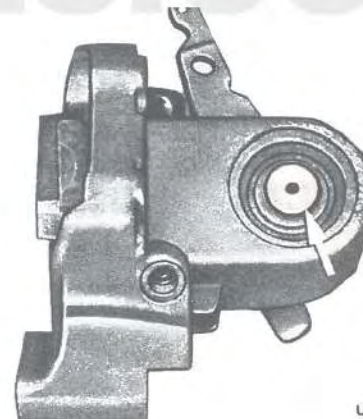
UR42-0451

2 Spray eccentric shaft with spray No. 3.



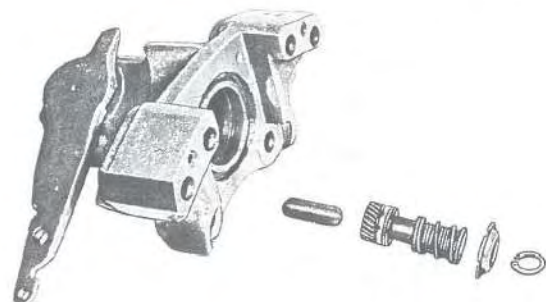
UR42-0748

3 Slide in actuating lever (25).



UR42-0749/1

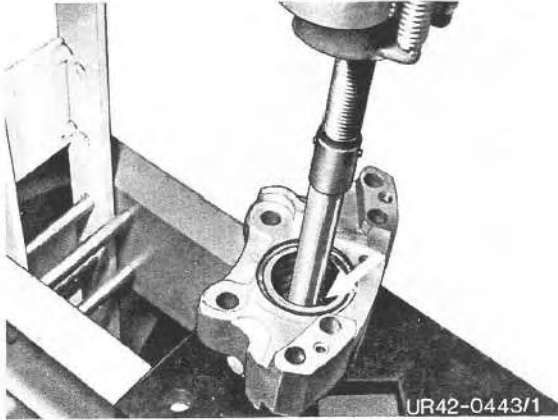
4 Install plunger (24), drive spindle (17), pressure spring (14) and spring plate (11).



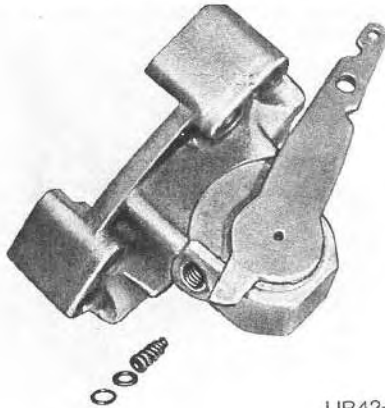
UR42-0450

42.11 Disassembly and Assembly of Rear Axle Fixed Caliper

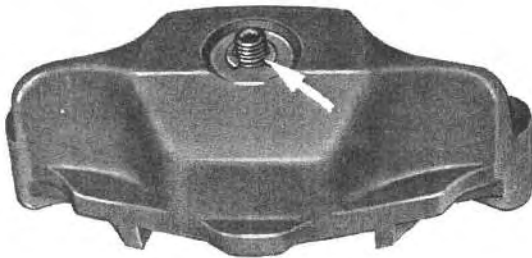
747.111/113



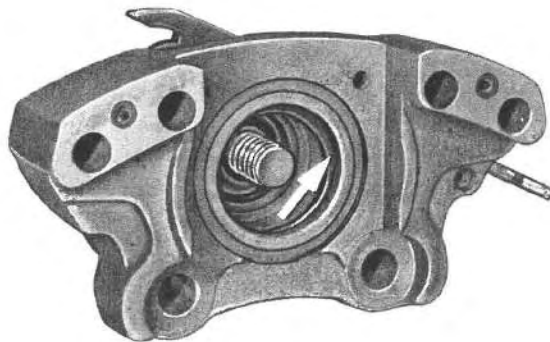
5 Compress spring (14) using special tool No. 14 and install retaining ring (12).



6 Fit restoring shaft (23) and thrust washer (21).

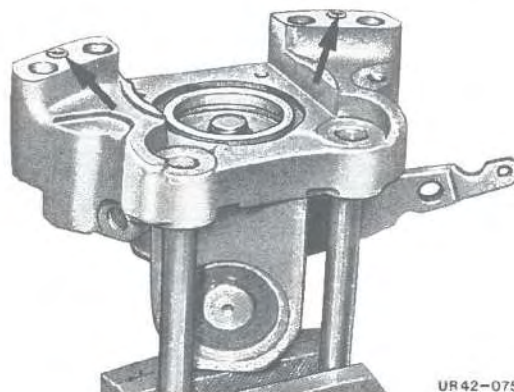


7 Install adjustment spindle (5).



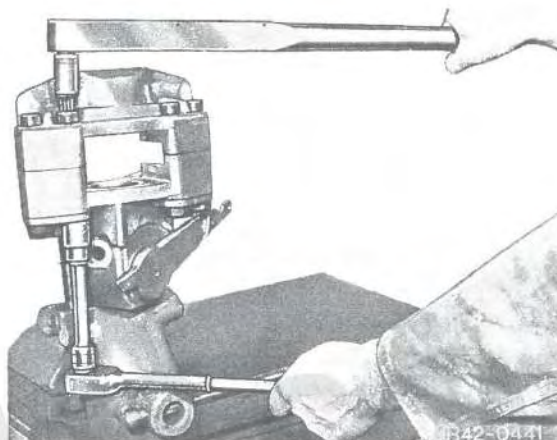
8 Spray sealing rings (7) with spray No. 3 and install.

- 9 Insert new sealing rings in flange section of housing.



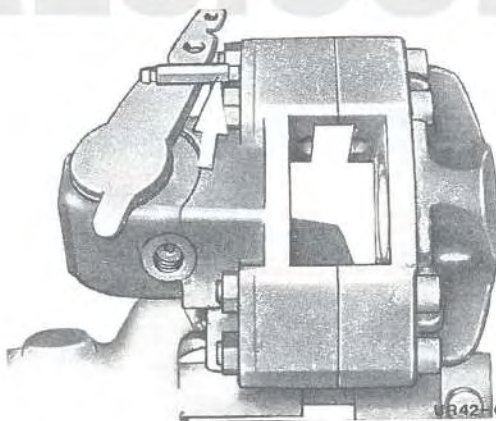
UR42-0751/1

- 10 Assemble both halves of caliper using new necked-down bolts and tighten with special tool No. 13. Tightening torque, see 1.3/1.



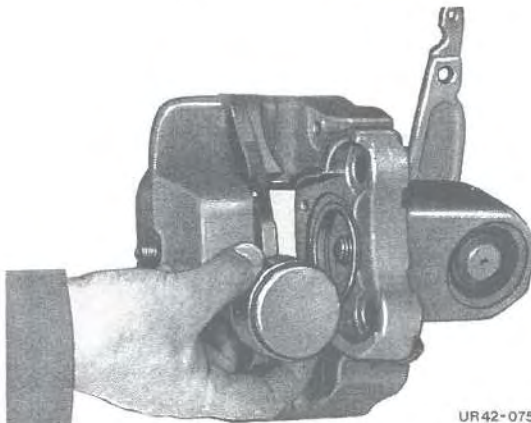
UR42-0441

- 11 Screw in bleeder valve. Tightening torque, see 1.3/1.



UR42-0744/1

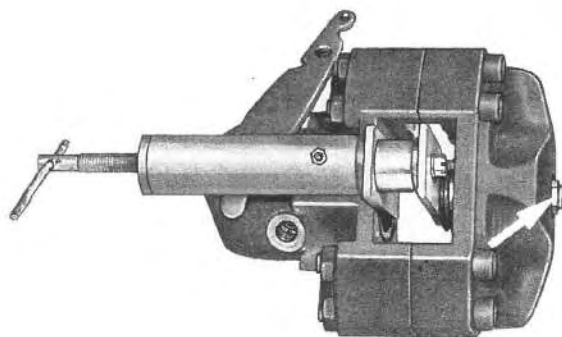
- 12 Coat piston (6) and cylinder bores with paste No. 2.



UR42-0752

42.11 Disassembly and Assembly of Rear Axle Fixed Caliper

747.111/113

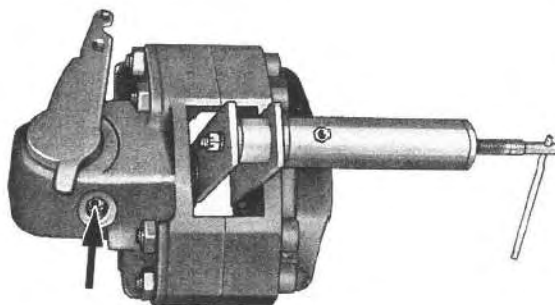


UR 42-0753/1

13 Fit piston (6) on the cover side by turning the adjustment spindle (5) such that the groove in the piston is still exposed.

Direction of rotation, see 1.5/1.

Note: Maintain pressure applied to piston with special tool No. 12.

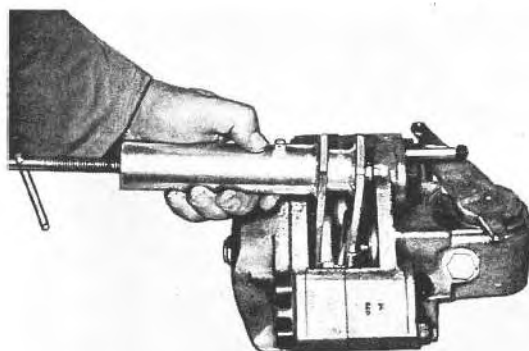


UR 42-0754/1

14 Install piston (6) on flange side by turning the restoring shaft (23) so that the groove in the piston is still exposed.

Direction of rotation, see 1.5/1.

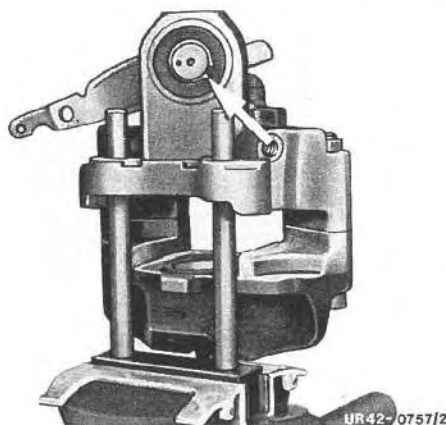
Note: Maintain pressure applied to piston using special tool No. 12.



UR42-0288

15 Half fill dust caps (10) with paste No. 2 and press on to shoulder of cylinder bore using special tools No. 8, No. 10 and No. 12.

For fixed calipers **with** hole in the eccentric shaft.

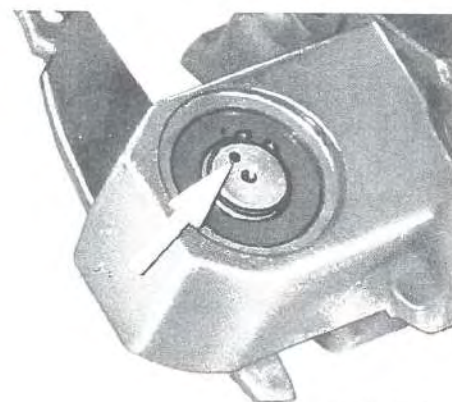


UR 42-0757/2

16 Fit support ring (28) and retaining ring (27).

17 Fill with special fluid No. 5 through hole in eccentric shaft.

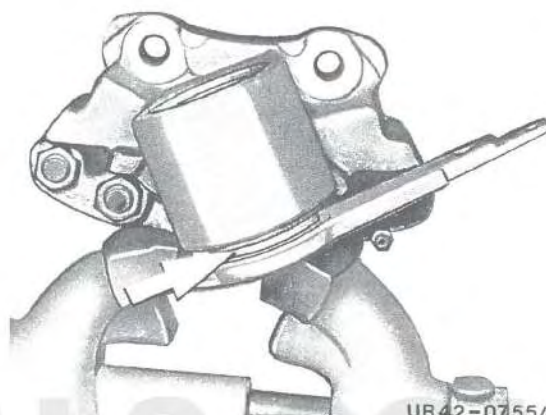
18 Continue to assemble as described from step No. 22.



UR42-0448/2

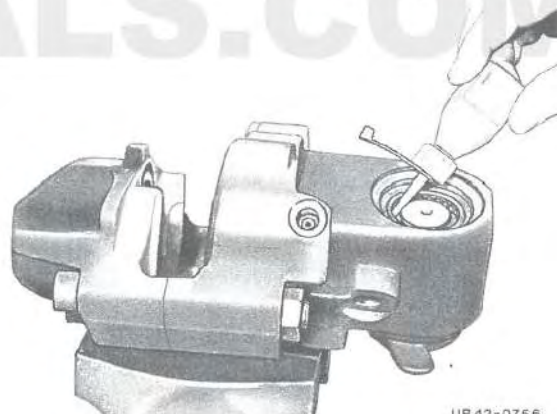
For calipers **without** hole in eccentric shaft

19 Slightly withdraw eccentric shaft.



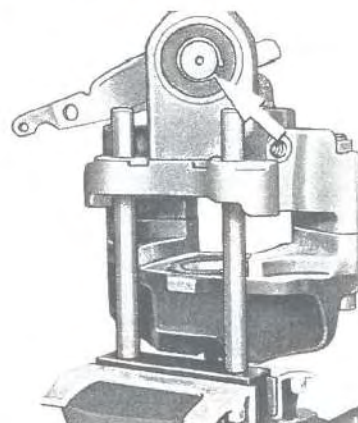
UR42-0755/1

20 Fill special fluid No. 5 between eccentric shaft and needle bearing.



UR42-0756

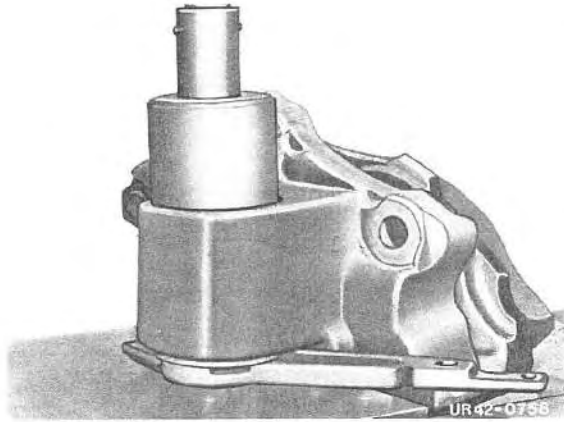
21 Fit support ring (28) and retaining ring (27).



UR42-0757/1

42.11 Diassembly and Assembly of Rear Axle Caliper

747.111/113



22 Press in new end disk using special tool No. 16.

23 Attach fixed caliper, see 35.6-6.1/6.

24 Install brake pads, see 3.1/3.

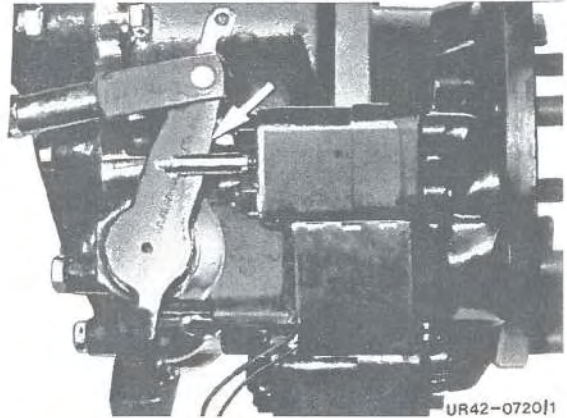
25 Adjust clearance, see 3.3/1.

26 Bleed brake system, see 5.1/1.

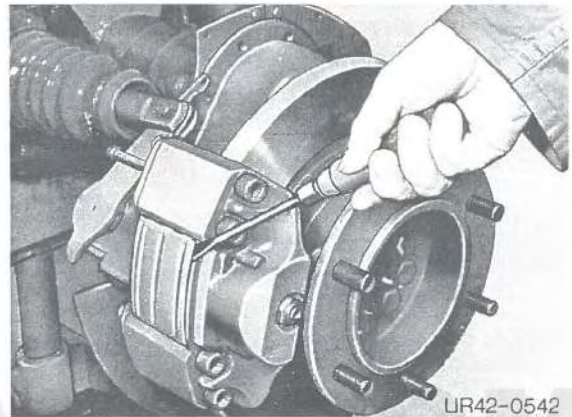
Adjusting

- 1 Release parking brake.

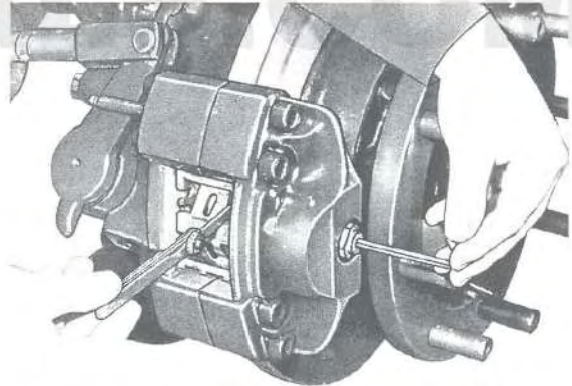
Note: The actuating lever must rest on the stop.



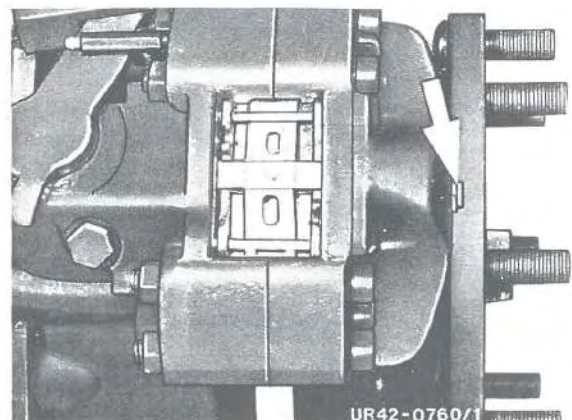
- 2 Remove guard plate.
Valid for brake lining wear indicator.



- 3 Loosen counternut on cover side, adjust clearance.
For clearance, see 1.1/1.
Direction of rotation, see 1.5/1.

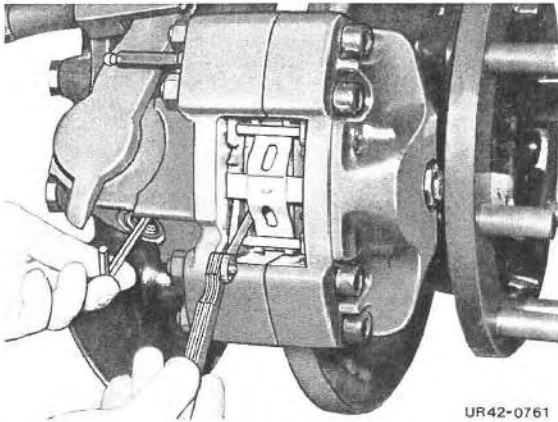


- 4 Tighten counternuts.
Tightening torque, see 1.3/1.

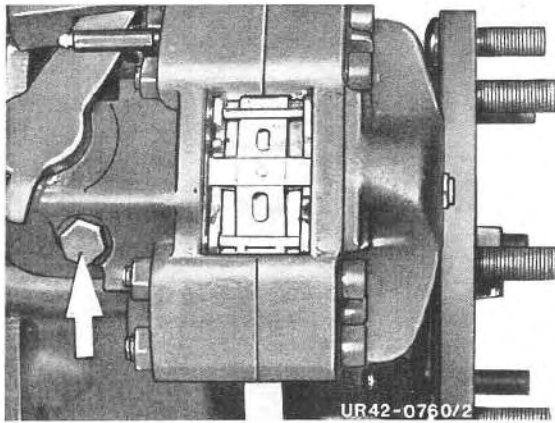


42.11 Adjusting Clearance in Rear Axle Fixed Caliper

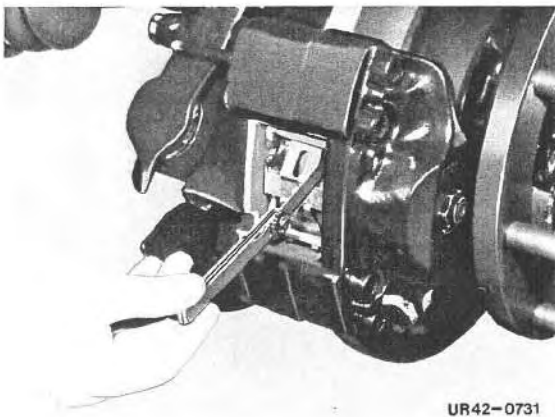
747.111/113



- 5 Unscrew screw plug on flange side, adjust clearance.
For clearance, see 1.1/1.
Direction of rotation, see 1.5/1.



- 6 Screw in screw plug.
Tightening torque, see 1.3/1.

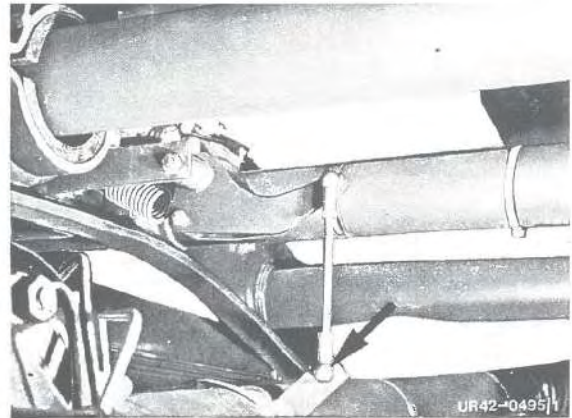


- 7 Actuate service brake and parking brake several times, check clearance several times.

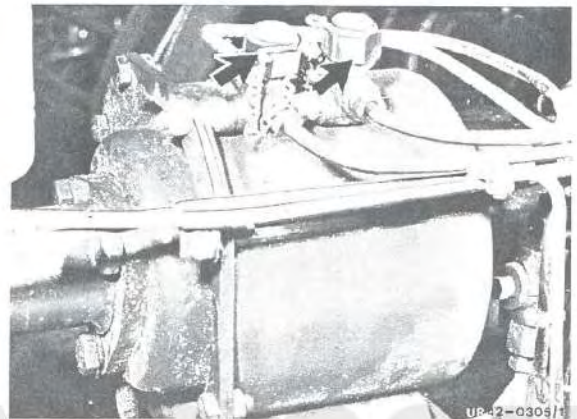
- 8 Install cover plate.
Valid for brake lining wear indicator.

Checking and Adjusting Failsafe System

1 Detach actuating linkage at torque tube.

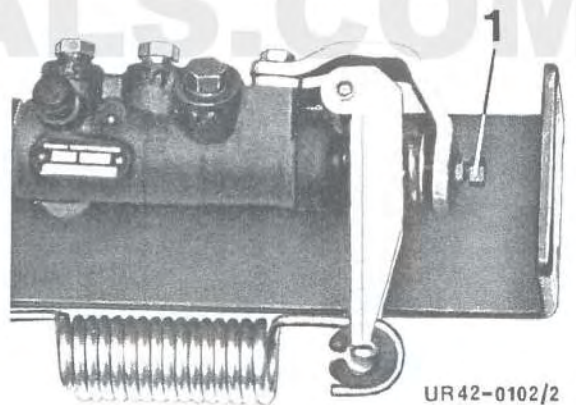


2 Connect a pressure gauge of the tester (special tool No. 2) to each of the two test connections upstream and downstream of the ALB modulator.



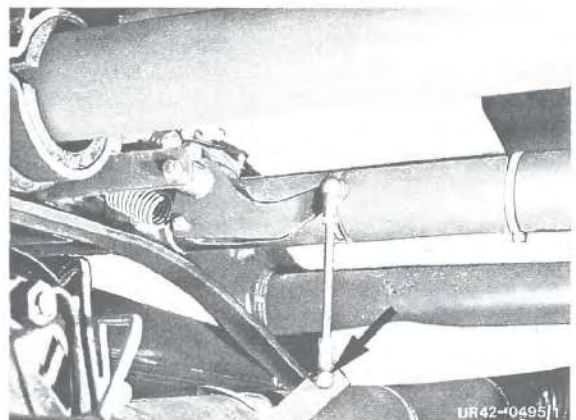
3 Slowly depress the brake pedal. The pressure indicated by both pressure gauges rises uniformly to the failsafe setting of 26 to 30 bar, from this point, the pressure in the modulated circuit must no longer rise.

4 If necessary, adjust the failsafe pressure of 26 to 30 bar using the adjusting screw (1).



5 Attach the actuating linkage to the torque tube.

6 Remove both pressure gauges and close off the test connections.



Checking and Adjusting the Load-Dependent Modulation

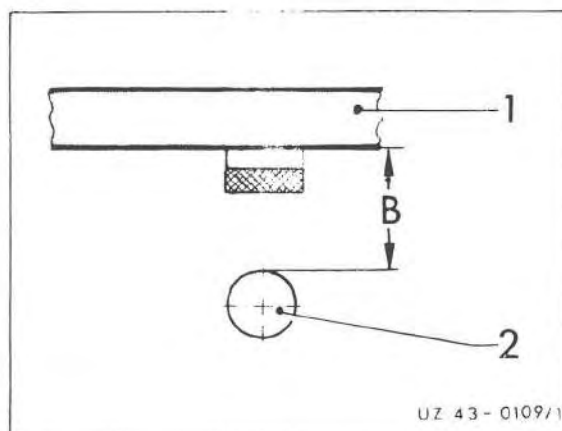
- 1 Check failsafe system and adjust if necessary, see 4.1/1.
- 2 Check modulator linkage for wear and that it is correctly secured.

To Chassis End No.

435.115 (without SA 35 988) : 001 771

453.115 (with SA 35 988) : 001 719

- 3 Measure deflection path (distance "B") between lower edge of frame (1) and axial bridge (2).



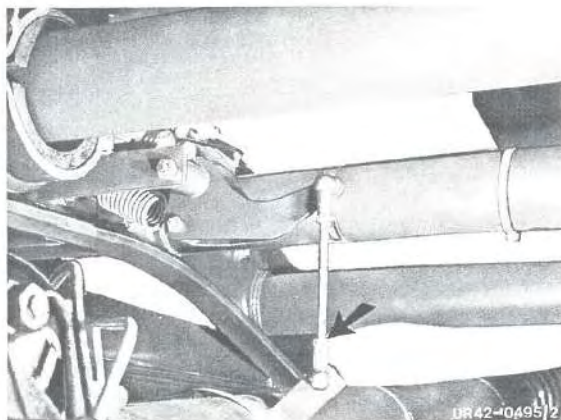
From Chassis End No.

435.115 (without SA 35 988) : 001 772

453.115 (with SA 35 988) : 001 719

435.117 from start of production

- 4 Determine rear axle load.
- 5 Fully depress brake pedal until the brake pressure of the unmodulated circuit reaches approx. 140 bar. The brake pressure in the modulated circuit must stop within the specified pressure range (see ALB diagram).



6 If necessary, correct modulated pressure by lengthening or shortening the linkage.

Note: Shortening the linkage reduces the pressure. Lengthening the linkage increases the pressure.

7 Unscrew both pressure gauges and close off the test connections.

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Directions for Bleeding

The brake system must be bled after any repair job that has involved the opening of the closed hydraulic system, or when the brake pedal feels soft and spongy. A variety of special-purpose fixtures, e.g. the ARC 50 unit or the ATE filler/bleeders, are commercially available for this purpose. Please remember to follow the instructions supplied with these units.

Brake fluid that has been bled off must not be put back because it may contain foreign matter which will then enter the brake system. Besides, brake fluid, being highly hygroscopic, steadily absorbs moisture from the atmosphere and gradually shows a reduced boiling point. As a result, vapour bubbles may form in the brake system under extreme conditions.

Brake fluid contains constituents which will dissolve paint. Therefore keep well away from vehicle paintwork.

While bleeding is in progress, ensure that bottom of refilling container is permanently covered by brake fluid. On no account must the compensating port be free from fluid because air will otherwise be aspirated when brake pedal is pressed, so rendering the entire operation futile.

Also make sure that the bleed vessel is held high enough to keep level on vessel higher than outlet from bleeder valve. Bleeding is complete when bubble-free brake fluid runs through the bleeder hose.

Having completed the entire operation, top up reservoir to mark "maximum" with brake fluid.

Note: If you bleed the system by "pumping" the brake pedal, remember to close the corresponding bleed screw each time before allowing the brake pedal to return. This serves to stop air from being aspirated through the bleed screw hole.

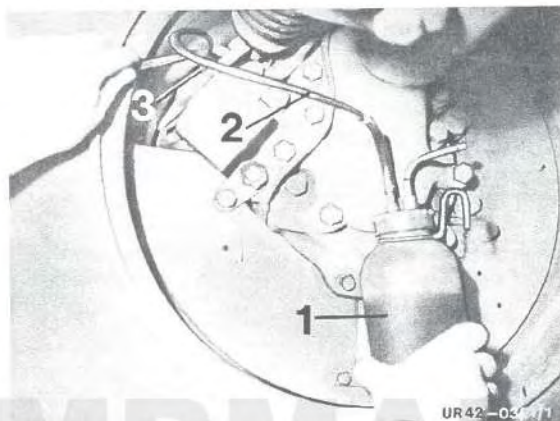
42.11 Bleeding the Brake System



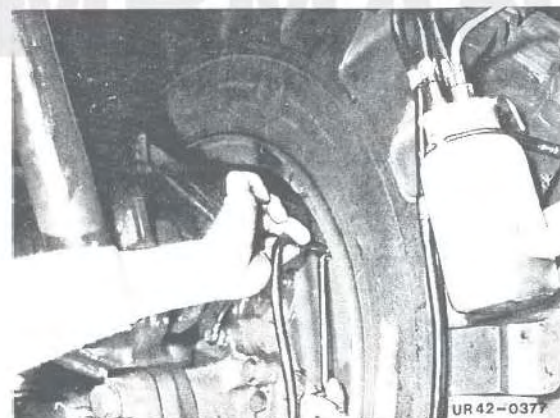
Bleeding

1 Prepare bleeder for use in accordance with manufacturer's specifications.

2 Unscrew caps on both reservoirs and connect special tools No. 2 and No. 3 to the bleeder.



3 Open bleeder screws one after the other on the master brake cylinder, ALB modulator and on the fixed calipers by 1 to 2 turns and close only when bubble-free brake fluid is obtained at the outlet.



4 Check that brake system functions correctly.

5 Detach bleeder.

Brake Lining Wear Measurement

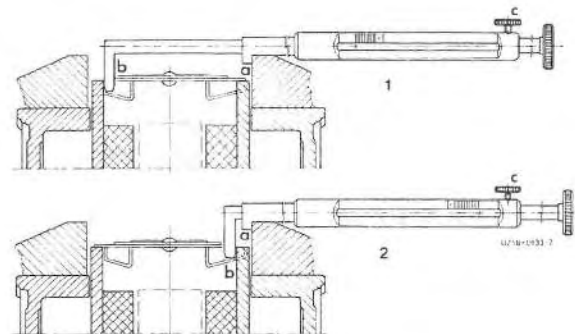
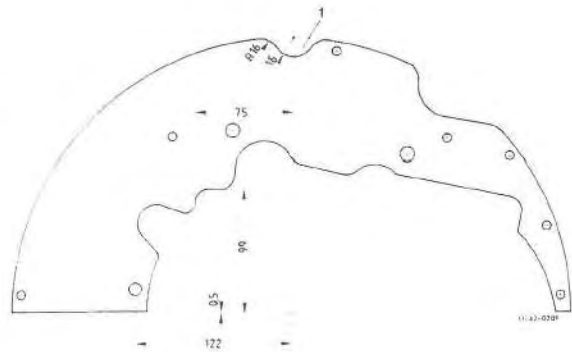
Note: Special tool No. 1 provides the possibility to check the thickness of the brake lining without removing the wheels provided no cover plates are fitted.

The shaft cover plates should be removed for fixed calipers **without** a brake lining wear indicator.

Mechanical wear measurement cannot be carried out on brake calipers with a brake lining wear indicator without removing the shaft cover plates. It may be possible to determine the thickness from the free caliper.

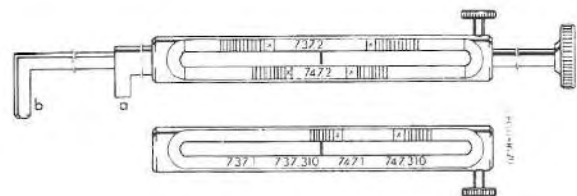
In order to ensure correct use of the special tool, a check must be made as to whether a recess which enables the special tool to be inserted between the wheel rim and the fixed caliper is provided on the brake backplate. If this is not the case, the recess should be provided in the brake backplates (1).

- 1 Using special tool No. 1 attach the hook "a" to the lower section of the shaft working from the inside of the wheel and place test piece "b" on the brake pad. Check brake lining thickness on cover side (1). Check brake lining thickness on flange side (2).



- 2 Tighten knurled screw "c" and read the brake lining wear off the scale. The brake pad must be renewed if the pointer is in the red area "x" (on the test gauge).

Note: When measuring, ensure the correct scale is used on the test gauge agreeing with the relevant axle model designation.



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Installation Survey

Chassis Model	Sales designation	Axles Model	Brake system ¹⁾ Tractor
435.110	U 1700	737.202	Dual-circuit hydraulic system and dual-circuit air brake system Disk brake ALB at rear axle ²⁾ Spring parking brake
435.111	U 1700 L	747.204	
435.113	U 1700 L/ 38	737.202 747.206	
435.160	Kit TM 170	737.232 747.231	

1) For pneumatic section, refer to group 43.14 - 1.1/1

2) ALB at rear axle and half front axle to chassis end No.

435.110: 086 854

435.111: 086 854

435.113: 084 641

Technical Data

Front and Rear Axle

Brake Disk

Outside diameter	mm	<u>380,00</u> 378,80
Inside diameter	mm	<u>145,06</u> 145,00
Thickness	mm	<u>30,00</u> 29,90
Permissible wear limit	mm	28,0
Permissible runout	mm	0,1

Brake Pads

Pad size when new	mm	10,0
Minimum pad size	mm	2,0
Pad, quality		Textar T 269

Fixed caliper

Clearance	mm	0,15 ± 0,05
-----------	----	-------------

Special Tools

Cons. No.	Tool	Number	Tool Set
1	Pin wrench 17 mm, 3/4 inch square	001 589 69 09 00	B
2	Test gauge for brake pads	440 589 00 19 00	B
3	Tester	001 589 64 21 00	B
4	Cylinder bore sealing plate	425 589 00 25 00	B
5	Twin hose for bleeder	000 589 02 25 00	B
6	Angled connection for bleeder	000 589 03 25 00	B
7	Counter support for extracting brake pads with special tool No. 5	000 589 34 33 00	B
8	Extractor for brake pads	001 589 43 33 00	B
9	Piston turning tool	000 589 50 37 00	B
10	Holder for installer	000 589 00 40 00	B
11	Piston resetting tool with tool No. 6 and No. 9 or No. 10 and No. 11	000 589 52 43 00	B
12	Installer for backplate diameter 48	000 589 53 43 00	B
13	Installer for backplate diameter 57 (only front axle)	000 589 54 43 00	B
14	Installer for protective caps diameter 48	000 589 55 43 00	B
15	Installer for dust caps diameter 57	000 589 56 43 00	B

Tightening Torques

Location		Thread	Nm
Wheel nuts		M 20 × 1,5	400
Fixed caliper on wheel hub drive housing		M 20 × 1,5	500
Fixed caliper	Conternut (cover side)	M 14 × 1	40
	Screw plug (flange side)		16 – 19
	Bleeder valve		10 – 14
	Threaded sleeve		25 – 28

Necessary Materials

Cons. No.	Designation	Part No.
1	Industrial alcohol	Commercially available
2	Lubricating grease	Commercially available
3	ATE brake fitting spray	Commercially available
4	ATE brake cylinder paste	Commercially available